

## MIDLAND, TX, USA

WMO: 722324

Lat: 32.036N

Lon: 102.101W

Elev: 854

StdP: 91.47

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03071

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -5.1       | -2.7 | -16.1                         | 1.0 | 7.3  | -13.9 | 1.2 | 8.6  | 13.0                  | 15.0 | 11.2 | 12.8 | 3.1                   | 40   | 0.517 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.9                   | 39.2            | 19.3 | 37.9 | 19.5 | 36.7 | 19.5 | 22.9                | 30.6 | 22.4 | 30.3 | 21.9 | 30.0 | 4.7                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.2                            | 17.6 | 25.0 | 20.5 | 16.8 | 24.9 | 20.0 | 16.3 | 24.9 | 72.0          | 30.4 | 70.1 | 30.1 | 68.5 | 29.9 | 25.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 10.8              | 9.2  | 8.3 | -8.5                       | 41.4 | 2.5                | 1.9 | -10.3                                              | 42.8 | -11.8      | 43.9 | -13.2      | 44.9 | -15.0      | 46.3 |    |    |
|                   |      |     | -9.7                       | 23.6 | 2.3                | 0.8 | -11.4                                              | 24.2 | -12.7      | 24.7 | -14.0      | 25.2 | -15.7      | 25.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 19.3   | 8.2   | 10.7  | 15.3  | 19.8  | 24.0  | 28.7  | 29.1  | 28.8  | 25.0  | 19.2  | 13.4  | 8.6   |
|                                                                   | DBStd       | 8.81   | 5.10  | 5.91  | 5.30  | 4.67  | 4.58  | 2.86  | 2.52  | 2.68  | 3.28  | 4.93  | 4.93  | 5.05  |
|                                                                   | HDD10.0     | 288    | 96    | 58    | 17    | 2     | 0     | 0     | 0     | 0     | 0     | 4     | 24    | 87    |
|                                                                   | HDD18.3     | 1217   | 316   | 219   | 121   | 38    | 10    | 0     | 0     | 0     | 2     | 51    | 158   | 303   |
|                                                                   | CDD10.0     | 3665   | 39    | 76    | 181   | 296   | 433   | 559   | 592   | 584   | 450   | 287   | 125   | 43    |
|                                                                   | CDD18.3     | 1554   | 0     | 4     | 27    | 82    | 185   | 309   | 333   | 326   | 202   | 76    | 9     | 1     |
|                                                                   | CDH23.3     | 19236  | 16    | 92    | 376   | 1000  | 2205  | 4027  | 4347  | 4178  | 2117  | 743   | 117   | 19    |
|                                                                   | CDH26.7     | 10149  | 1     | 18    | 115   | 423   | 1158  | 2324  | 2466  | 2358  | 998   | 268   | 21    | 1     |
| Wind                                                              | WSAvg       | 4.1    | 3.8   | 4.2   | 4.6   | 5.0   | 4.7   | 4.8   | 4.0   | 3.6   | 3.6   | 3.8   | 3.6   | 3.7   |
| Precipitation                                                     | PrecAvg     | 356    | 16    | 22    | 20    | 21    | 42    | 45    | 45    | 45    | 43    | 33    | 21    | 13    |
|                                                                   | PrecMax     | 599    | 54    | 65    | 62    | 74    | 180   | 94    | 171   | 91    | 115   | 111   | 139   | 69    |
|                                                                   | PrecMin     | 163    | 0     | 0     | 2     | 0     | 3     | 0     | 1     | 4     | 1     | 1     | 0     | 0     |
|                                                                   | PrecStd     | 120    | 16    | 19    | 16    | 22    | 47    | 27    | 45    | 28    | 34    | 33    | 31    | 17    |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 25.3  | 29.0  | 32.3  | 35.2  | 40.0  | 41.7  | 40.3  | 40.0  | 37.1  | 33.8  | 29.0  | 25.8  |
|                                                                   |             | MCWB   | 11.2  | 12.6  | 13.7  | 14.7  | 17.0  | 18.8  | 20.6  | 19.9  | 18.9  | 17.9  | 13.9  | 11.7  |
|                                                                   | 2%          | DB     | 22.6  | 26.1  | 29.9  | 33.1  | 37.2  | 39.6  | 38.4  | 38.4  | 35.3  | 32.0  | 26.4  | 22.7  |
|                                                                   |             | MCWB   | 10.1  | 11.7  | 13.1  | 14.5  | 16.0  | 18.6  | 20.2  | 20.1  | 19.1  | 17.3  | 13.9  | 10.9  |
|                                                                   | 5%          | DB     | 20.1  | 23.7  | 27.6  | 31.4  | 35.1  | 37.7  | 37.2  | 37.2  | 33.9  | 30.0  | 24.3  | 20.1  |
|                                                                   |             | MCWB   | 9.2   | 11.2  | 12.7  | 14.3  | 16.5  | 19.3  | 20.3  | 20.3  | 18.9  | 17.3  | 13.0  | 9.9   |
|                                                                   | 10%         | DB     | 17.4  | 21.0  | 25.2  | 29.2  | 32.8  | 36.1  | 35.8  | 35.8  | 32.1  | 27.7  | 22.3  | 17.3  |
|                                                                   |             | MCWB   | 8.6   | 10.0  | 12.4  | 14.2  | 16.6  | 19.6  | 20.2  | 20.3  | 19.2  | 16.7  | 12.5  | 9.2   |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 14.5  | 15.6  | 17.9  | 20.0  | 22.5  | 23.4  | 23.6  | 23.3  | 22.9  | 22.3  | 18.0  | 15.4  |
|                                                                   |             | MCDB   | 18.4  | 21.4  | 23.5  | 26.7  | 28.3  | 31.5  | 30.1  | 31.3  | 29.4  | 28.8  | 21.3  | 19.0  |
|                                                                   | 2%          | WB     | 12.1  | 14.2  | 16.8  | 18.9  | 21.3  | 22.7  | 22.6  | 22.6  | 22.0  | 20.8  | 16.5  | 13.2  |
|                                                                   |             | MCDB   | 18.4  | 20.7  | 22.4  | 24.7  | 27.7  | 31.1  | 30.0  | 31.1  | 28.7  | 25.8  | 21.1  | 17.4  |
|                                                                   | 5%          | WB     | 10.4  | 12.6  | 15.8  | 17.8  | 20.6  | 22.1  | 22.2  | 22.1  | 21.5  | 20.0  | 15.4  | 11.5  |
|                                                                   |             | MCDB   | 18.4  | 20.6  | 21.7  | 23.8  | 27.1  | 30.2  | 30.1  | 30.8  | 28.0  | 25.5  | 20.3  | 18.1  |
|                                                                   | 10%         | WB     | 8.9   | 11.1  | 14.7  | 16.8  | 19.8  | 21.6  | 21.7  | 21.7  | 21.0  | 19.0  | 14.1  | 9.9   |
|                                                                   |             | MCDB   | 16.7  | 19.5  | 21.3  | 23.6  | 26.8  | 29.8  | 30.2  | 30.4  | 27.2  | 24.7  | 20.3  | 17.2  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 13.2  | 13.9  | 14.4  | 14.7  | 13.5  | 13.2  | 11.9  | 12.0  | 11.7  | 12.8  | 13.4  | 12.6  |
|                                                                   |             | MCDBR  | 17.2  | 17.8  | 17.1  | 17.1  | 16.0  | 15.2  | 13.6  | 13.5  | 13.9  | 14.7  | 15.7  | 16.8  |
|                                                                   |             | MCWBR  | 7.9   | 7.9   | 6.7   | 6.3   | 5.7   | 3.9   | 3.6   | 3.2   | 3.9   | 5.4   | 6.7   | 8.1   |
|                                                                   | 5% WB       | MCDBR  | 15.1  | 15.5  | 13.6  | 13.3  | 13.2  | 12.9  | 11.6  | 11.8  | 10.7  | 11.5  | 12.3  | 13.8  |
|                                                                   |             | MCWBR  | 7.9   | 8.1   | 6.9   | 6.4   | 5.7   | 4.0   | 3.3   | 3.3   | 3.4   | 5.1   | 6.9   | 7.9   |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.266 | 0.277 | 0.297 | 0.333 | 0.350 | 0.379 | 0.396 | 0.373 | 0.354 | 0.316 | 0.288 | 0.270 |
|                                                                   | taud        |        | 2.575 | 2.545 | 2.502 | 2.396 | 2.415 | 2.378 | 2.377 | 2.413 | 2.434 | 2.524 | 2.536 | 2.561 |
|                                                                   | Ebn at Noon |        | 976   | 991   | 987   | 956   | 936   | 903   | 886   | 905   | 913   | 932   | 938   | 950   |
|                                                                   | Edh at Noon |        | 80    | 91    | 102   | 119   | 118   | 122   | 122   | 116   | 109   | 92    | 82    | 77    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 3.46  | 4.28  | 5.54  | 6.78  | 7.18  | 7.59  | 7.20  | 6.68  | 5.56  | 4.55  | 3.62  | 3.08  |
|                                                                   | RadStd      |        | 0.34  | 0.45  | 0.40  | 0.42  | 0.52  | 0.36  | 0.40  | 0.30  | 0.38  | 0.51  | 0.38  | 0.28  |

## Historical Trends

|                         |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                     | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| Station Only            | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (15 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +109    | +75     |

Nomenclature: See separate page